

5 Further kinematics	5.1	Kinematics of a particle moving in a straight line when the acceleration is a function of the displacement (x), or time (t) or velocity (v).	The setting up and solution of equations where $\frac{dv}{dt} = f(t) \text{ or } \frac{dv}{dt} = f(v),$ $v \frac{dv}{dx} = f(v) \text{ or } v \frac{dv}{dx} = f(x),$ $\frac{dx}{dt} = f(x) \text{ or } \frac{dx}{dt} = f(t)$ will be consistent with the level of calculus required in A level Mathematics and Further Mathematics.
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